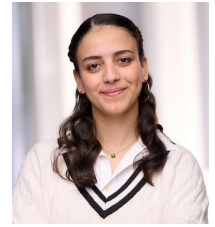


Hiba Assamaouat | Electrical Engineer

✉ ha1647@nyu.edu 📞 +212 682 867 960 [in hiba-assamaouat](#) 🌐 hiba-assamaouat.github.io



Electrical Engineer with a strong focus on optimization, organization, and project management. I seek a company that values integrity and ambition. Through my engineering experiences, I have acquired a unique perspective with regards to leadership and great attention to detail.

EXPERIENCE

Intern | Embedded Co-Design for Automotive Applications Mar. 2026 – Present
Institut National des Postes et Télécommunications (INPT) Rabat, Morocco

- Leading the end-to-end setup of a gem5 simulation environment to generate performance datasets for an ML model targeting Processing-in-Memory (PIM) co-design optimization for energy efficiency and compute performance.
- Benchmarking automotive Electronic Control Units (ECU) workloads across hardware families to produce labeled microarchitectural data, directly feeding upstream ML model training.
- Structured a multi-component simulation repository with version control, modular configuration, and documented workflows to ensure reproducibility and handoff-readiness within a PhD research team.

Project Assistant | Wireless Research Department Nov. 2024 – Jun. 2025
New York University Abu Dhabi Abu Dhabi, UAE

- Conducted a comprehensive review of ~10 papers on Synthetic Aperture Radar (SAR) image despeckling and super-resolution, analyzing deep learning approaches and architectural trends.
- Surveyed and assessed open-source and private SAR datasets for suitability in super-resolution tasks, contributing to dataset selection and experimental planning.
- Worked closely with a PhD researcher to study and compare candidate deep learning architectures, developing a solid conceptual understanding of model design and training workflows.
- Contributed to research discussions on model evaluation strategies and computational requirements for SAR super-resolution experiments on HPC systems.

Electrical Engineering Summer Intern Jul. 2023
Redal VEOLIA Rabat, Morocco

- Visited and studied multiple electrical substations across the city, gaining hands-on exposure to power distribution system design, operation, and maintenance practices.
- Worked with departmental leads and the dispatching team to understand coordination, monitoring, and decision-making in power distribution network management.
- Assisted technicians in troubleshooting and repairing power distribution components in both urban substations and remote rural areas, supporting field operations.

Research Assistant | Plastic Recycling Lab Oct. 2022 – May 2023
New York University Abu Dhabi Abu Dhabi, UAE

- Worked in a lab affiliated with the global Precious Plastic community, applying open-source principles in sustainable design and plastic recycling.
- Contributed to the design and early development of a plastic recycling system converting HDPE bottle caps collected on campus into 3D-printing filament.
- Collaborated with lab members to support progress and maintain alignment with research objectives and timelines.
- Led outreach activities, including on-campus tabling and lab tours, to raise awareness and encourage student participation in recycling initiatives.

EDUCATION

New York University Abu Dhabi (NYUAD), Abu Dhabi, UAE Aug. 2021 - May 2025
Bachelor of Science in Electrical Engineering

- Studies included a semester abroad in New York, a short-term stay at NYU Czech Republic, and a seminar in Paris
- **Relevant Coursework:** Electronics; Control Systems; Photonics; Communication Theory; Electromagnetics; Instrumentation; Embedded Systems; Object-Oriented Programming

Lycée Français International Théodore Monod (LFITM), Abu Dhabi, UAE Jun. 2021
French Baccalaureate, Summa cum laude ("Mention très bien")

RELEVANT SKILLS

Frameworks & Development Tools: Simulink, Altium Designer, KiCad, VS Code, Jupyter, PlatformIO, ArduinoIDE, Fusion360, Git, LTspice, PyTorch, Bash
Programming: Python, MATLAB, C/C++, HTML, CSS, JavaScript, VHDL
Software: Microsoft Office, L^AT_EX, iMovie, Notion, Procreate
Hardware: Soldering, Surface-Mounted Device (SMD) Soldering, Circuitry, Arduino, AdaFruit Circuit Playground, AtMega32u4
Languages: Arabic: *Native* | French: *Fluent* | English: *Fluent* | Spanish: *Intermediate*

ENGINEERING PROJECTS

AI-Powered Tech News Fetcher Mistral AI Online Hackathon	2026
Design and implementation of a full-stack news aggregation tool combining real-time article retrieval with AI-powered summarization. Built a Flask API backend coordinating NewsAPI and Mistral AI, with a vanilla JavaScript frontend for dynamic rendering. (Python, Flask, Mistral AI, NewsAPI, JavaScript, HTML/CSS)	
Multi-Stage Peltier Module (TEC) Refrigerator NYUAD	2025
(Team Lead) Design and implementation of a thermoelectric refrigeration system using multi-stage Peltier (TEC) modules instead of conventional compressor-based cooling. Development of control circuitry and PCB layout. (Arduino, KiCad)	
Electrocardiography (ECG) PCB Design NYUAD	2023
Design and fabrication of a printed circuit board for ECG signal acquisition, emphasizing low-noise analog front-end architecture and signal integrity optimization. (Altium Designer)	
Self-Balancing Robot Control System NYUAD	2024
Design and implementation of a two-wheel self-balancing robotic platform using PID-based control. Controller tuning to minimize oscillations and enhance dynamic stability. (MATLAB, Simulink)	
Parkinson’s Tremor Detection System NYUAD	2024
Design of a wearable microcontroller-based tremor detection system using accelerometer data. Implementation of band-pass filtering and frequency-domain calibration for tremor intensity estimation. (Arduino, Adafruit)	
End-to-End Wireless Communication System NYUAD	2024–2025
Design of a complete digital communication system from transmitter to receiver. Implementation of synchronization techniques for operation in noisy channels and reduction of bit error rate through advanced signal processing methods. (MATLAB, USRP)	
SIRD Epidemiological Model (RK4 Implementation) NYUAD	2023
Modeling of infectious disease propagation using a SIRD (Susceptible-Infected-Recovered-Deceased) framework. Implementation of a fourth-order Runge-Kutta solver with analysis of numerical stability and sensitivity to initial conditions. (MATLAB)	
Lagrangian-Based Recommendation System NYUAD	2023
Development of a recommendation system based on constrained optimization using Lagrangian formulation. Definition of cost functions and mitigation of user-item sparsity challenges. (MATLAB)	

**See all projects on my [personal page](#)*

EXTRACURRICULARS

Intercollegiate Soccer NYUAD Athletics	2021 - 2023
Took part in the NYUAD women’s soccer team.	
Indoor Rock Climbing NYUAD Athletics	2024 - 2025
Pursued independent training at the NYUAD indoor wall.	